Literature Review (2024-2026)

Ref. No.	Author name & Year	Publication name	Article Title	Observation	Limitations
[1]	Kumar et al. & 2026	IEEE / MIGT	Agentic AI Meets Edge Computing in Autonomous UAV Swarms	Proposed a hybrid edge-enabled LLM agentic AI framework (using U-Net segmentation, GPT-4.1 at EGS, and TinyLLaMA on UAVs) for wildfire search-and-rescue, achieving dynamic coverage and minimal mission completion times.	Onboard LLMs face computational and battery trade-offs; susceptible to LLM hallucination and inter-agent error propagation during dynamic mission planning.
[2]	Yu & Zhou & 2026	Elsevier / Digital Engineering	Distributed multi-agent proximal policy optimization for non-holonomic robot trajectory planning	Proposed DMAPPO, demonstrating key MAPPO mechanisms adaptable to drones: (1) Synchronous distributed training with a client-server parameter-averaging loop to eliminate policy staleness, (2) Composite reward design balancing goal arrival with continuous acceleration jerk penalties and graded proximity buffers, and (3) CTDE architecture to mitigate multi-agent non-stationarity.	Evaluated strictly on 2D ground AMRs using Ackermann kinematic constraints and planar Velocity Obstacles, requiring translation to 3D 6-DOF flight dynamics for drone deployment; assumes ideal communication without latency.

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[3]	Pal et al. & 2026	IEEE / AIEI	Multi-Agent Reinforcement Learning with Decentralized AI in Autonomous Drone Swarms	Combined YOLOv8 detection, DeepSORT tracking, and Multi-Agent PPO (in DroneSwarmEnv) to enable autonomous formation control and collision avoidance in drone swarms at 151.5 FPS.	Evaluated in semi-simulated environments that do not fully capture real-world conditions like wind turbulence, GPS drift, or communication delays.
[4]	Cha et al. & 2026	IEEE / WCNC	Reasoning Meets Adaptation: Bridging Global LLM Planning and Distributed UAV Control	Introduced a hierarchical FANET formation framework where an LLM provides global coordination points via obstacle relaxation, and distributed UAV local policies execute real-time collision-free routing.	Obstacle-relaxed LLM planning yields coordination points that may overlap or collide with obstacles, placing total reliance on local UAV policies for physical safety.
[5]	Sayyad et al. & 2025	IEEE / CNC	Simulation-Based Evaluation and Communication Strategies in Multi-Drone Systems	Evaluated Q-Learning and Deep Q-Network (DQN) algorithms in a multi-agent simulation using IR checkpoint localization for dynamic pathfinding and obstacle avoidance.	Validated on a small scale (2 UAVs) in a 2D simulation environment, limiting applicability to complex 3D real-world conditions with severe environmental noise.

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[6]	Hend Khalid et al.& 2026	IEEE	Explainable Multi-Agent Reinforcement Learning for Trust-Aware UAV Swarm Communication	Proposed an **EMARL** framework using **MADDPG** with trust-aware routing and explainable AI (SHAP, LIME) to improve secure and transparent UAV swarm communication.	Focuses on communication security only; does not address mission planning, task allocation, or formation control.
[7]	Xinxin Yuan et al., 2026	Applied Sciences (MDPI	A Multi-Agent Deep Reinforcement Learning Approach with Communication Resource Optimisation for Cooperative UAV Swarm Perception and Multi-Target Detection in Emergency Rescue	Proposed a **MAPPO-CTDE** framework with detection-aware observations, lightweight communication, and a composite reward to improve cooperative perception and multi-target search	Limited to emergency rescue scenarios and cooperative perception; does not address mission planning or heterogeneous UAVs.
[8]	Sun et al., 2024	ICAUS 2024 (Springer)	Research on the Model-Based Ad Hoc Teamwork Reinforcement	Proposed a **Model-Based Ad Hoc Teamwork RL** framework enabling UAVs to cooperate with previously	Focuses on navigation only; does not consider communication optimization,

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			Learning Method for Cooperative Navigation Task of Drone Swarm	unseen teammates for adaptive swarm navigation.	mission planning, or task allocation.
[9]	Dong et al. (2026)	IEEE	Collaborative Beamforming for Secure UAV Swarm Communications: An End-to-End MAPPO-Based Framework Against Mobile Eavesdroppers	Proposed a **MAPPO-based** framework that jointly optimizes UAV trajectories and collaborative beamforming to improve secure swarm communication against mobile eavesdroppers using the **CTDE** framework g.	Focuses on secure communication and trajectory optimization. Does not address mission planning, task allocation, or heterogeneous UAV coordination.

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[10]	Almutairi et al., 2025	Algorithms (MDPI)	Unmanned Aerial Vehicles (UAV) Networking Algorithms: Communication, Control, and AI-Based Approaches	Reviewed UAV networking, FANET, routing, control architectures, and AI-based techniques for autonomous UAV swarms, highlighting future research challenges.	does not propose or evaluate a new AI coordination framework.
11	Yan et al., 2024	Elsevier	Cooperative Task Allocation with Simultaneous Arrival and Resource Constraint for Multi-UAV Using a Genetic Algorithm	Proposed a modified **Genetic Algorithm (GA)** for cooperative multi-UAV task allocation and path planning considering simultaneous arrival, resource constraints, and deadlock avoidance to improve mission coordination.	Uses a Genetic Algorithm for optimization rather than an AI-enabled multi-agent framework, limiting autonomous collaboration, adaptive mission execution, and dynamic task reallocation in coordinated UAS operations.

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12	**Wang et al., 2024**	Elsevier	An Evolutionary Multi-Agent Reinforcement Learning Algorithm for Multi-UAV Air Combat	Proposed an **Evolutionary Multi-Agent Reinforcement Learning (E-MATD3)** framework that combines evolutionary algorithms, attention mechanisms, and CTDE to improve cooperative decision-making, scalability, and tactical coordination in multi-UAV air combat.	Focuses on multi-UAV air combat and lacks dynamic task allocation, heterogeneous UAV coordination, and general coordinated UAS functions.
13	**Xu et al., 2025**	Elsevier	Multi-objective Evolutionary Algorithm with Two Balancing Mechanisms for Heterogeneous UAV Swarm Path Planning	Proposed a **Heterogeneous Multi-Objective Evolutionary Algorithm (HMOEA)** with two balancing mechanisms to optimize heterogeneous UAV swarm path planning by jointly minimizing flight distance, flight time, and energy consumption while improving convergence and solution diversity.	Focuses on path planning optimization and does not address AI-enabled multi-agent coordination, dynamic task allocation, or autonomous decision-making for coordinated UAS functions.

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14	Yang et al., 2026	IEEE Communications Surveys & Tutorials	Advancing Multi-Robot Networks via MLLM-Driven Sensing, Communication, and Computation: A Comprehensive Survey	Reviewed MLLM-driven sensing, communication, and computation frameworks for coordinated multi-robot networks.	AI methods for coordinating multiple robots while adapting to changing tasks and environments are still limited.
15	Gao et al., 2024	IEEE Internet of Things Journal	MARL-Based UAV Trajectory and Beamforming Optimization for ISAC System	Proposed an **HR-MAPPO** framework for joint UAV trajectory and beamforming optimization in multi-UAV ISAC systems.	Dynamic mission planning and task allocation for coordinated multi-UAV operations remain unexplored.

RESEARCH GAP

- Lack of dynamic mission planning and task allocation
- Limited coordination of heterogeneous UAVs

- Poor adaptability to dynamic real-world environments
- Fragmented solutions for coordinated UAS functions
- Autonomous decentralized multi-agent decision-making remains underdeveloped.